Day 08: Networking Challenge

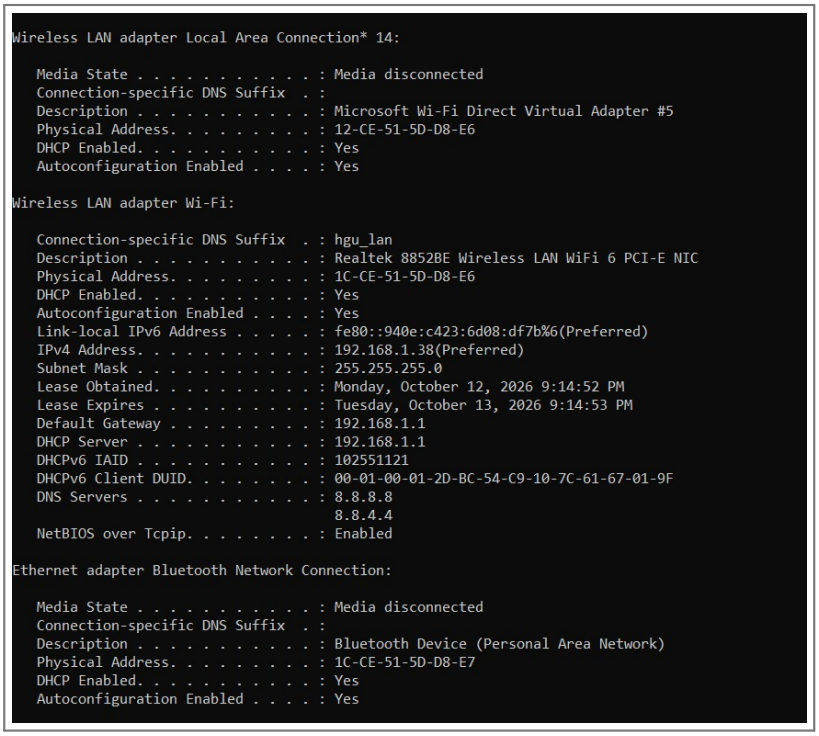
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Task 1: Network Identity

Details recorded from **ipconfig /all** on the active Wi-Fi adapter.

Item	Value
IPv4 Address	192.168.1.38
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	8.8.8.8 (primary), 8.8.4.4 (secondary)
MAC / Physical Address	1C-CE-51-5D-D8-E6
DHCP Enabled	Yes

Figure 1: ipconfig /all - Wi-Fi Adapter



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Wireless LAN adapter Local Area Connection* 14:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #5
   Physical Address. . . . . . . . . : 12-CE-51-5D-D8-E6
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : hgu_lan
   Description . . . . . . . . . . . : Realtek 8852BE Wireless LAN WiFi 6 PCI-E NIC
   Physical Address. . . . . . . . . : 1C-CE-51-5D-D8-E6
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::940e:c423:6d08:df7b%6(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.1.38(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Monday, October 12, 2026 9:14:52 PM
   Lease Expires . . . . . . . . . . : Tuesday, October 13, 2026 9:14:53 PM
   Default Gateway . . . . . . . . . : 192.168.1.1
   DHCP Server . . . . . . . . . . . : 192.168.1.1
   DHCPv6 IAID . . . . . . . . . . . : 102551121
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2D-BC-54-C9-10-7C-61-67-01-9F
   DNS Servers . . . . . . . . . . . : 8.8.8.8
                                       8.8.4.4
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Bluetooth Device (Personal Area Network)
   Physical Address. . . . . . . . . : 1C-CE-51-5D-D8-E7
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
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The Wi-Fi adapter shows the IPv4 address, subnet mask, default gateway, DNS servers, physical address and DHCP status listed above.

Task 4: DNS Investigation

Command used: **nslookup google.com**

Question	Answer
Which DNS server responded?	192.168.100.1 (port 53)
Which IP address was returned?	192.178.193.138, 192.178.193.102, 192.178.193.100, 192.178.193.113, 192.178.193.139, 192.178.193.101
What is the purpose of DNS?	DNS translates domain names such as google.com into IP addresses so devices can find and connect to each other.

Figure 2: nslookup google.com



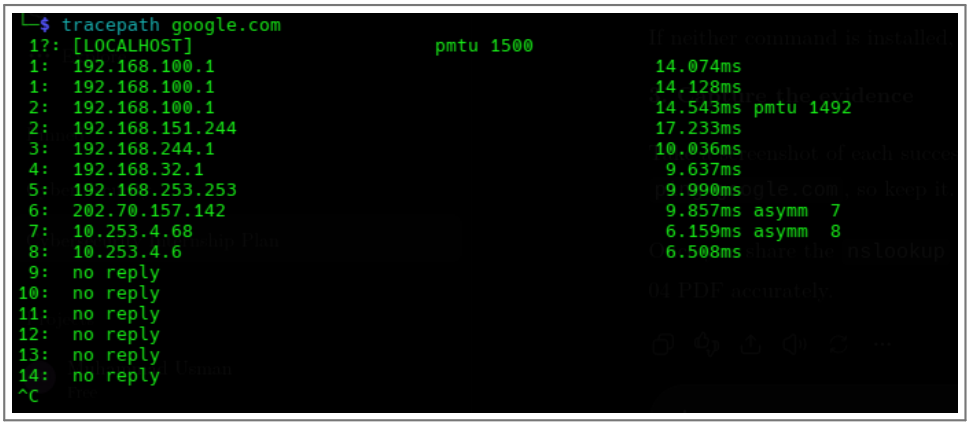
The DNS server 192.168.100.1 answered on port 53. The answer is non-authoritative and returned six IP addresses for google.com.

Task 5: Network Path Investigation

Command used: **tracepath google.com** (the Kali Linux equivalent of tracert)

Question	Answer
How many visible hops appeared?	14 hops were listed (hops 1 to 14). Hops 1 to 8 replied and hops 9 to 14 showed no reply.
What was the first hop?	192.168.100.1
What device commonly represents the first hop?	The default gateway, usually the local router.
What is traceroute used for?	It shows the path packets take to a destination, router by router, and helps find where delays or failures happen.

Figure 3: tracepath google.com



The path starts at the local gateway 192.168.100.1 and passes through private and ISP addresses until hop 8 (10.253.4.6). Hops 9 to 14 show no reply.